LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Personalized Meal & Diet Subscription Manager

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1. Architecture Selection

We chose Layered Architecture for the Personalized Meal & Diet Subscription Manager.

Reason 1 – Separation of Responsibilities: The system has distinct responsibilities such as managing dietary profiles, generating and validating meal plans, reviewing meal plans, and managing subscriptions and deliveries. Layered Architecture separates these responsibilities into Presentation, Business, and Data layers, making the system easier to understand and maintain.

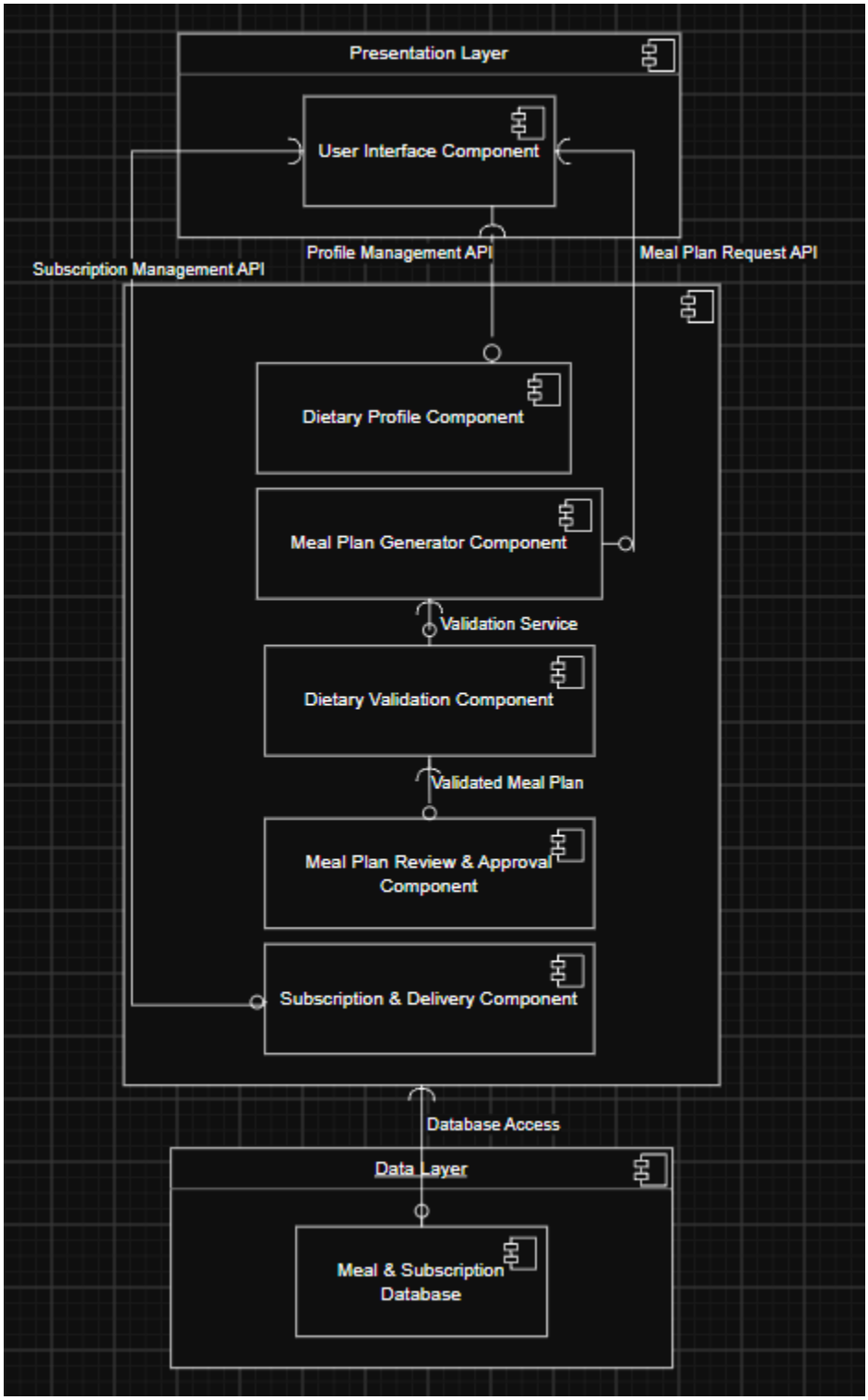
Reason 2 – Maintainability and Modularity: The architecture allows individual business components such as the Dietary Profile, Meal Plan Generator, Dietary Validation, Meal Plan Review & Approval, and Subscription & Delivery components to be modified independently while maintaining clear interfaces between layers.

2. Security Advantage

The layered architecture helps restrict direct access to the database. User requests pass through the Presentation and Business layers before reaching the Data Layer. This separation allows authentication, authorization, and input validation to be implemented in the appropriate layers and reduces direct exposure of stored dietary and subscription information.

3. Performance Benefit

Separating the system into components allows frequently used business operations, such as meal-plan generation and dietary validation, to be optimized independently. Efficient database access and component-level processing can reduce unnecessary processing and improve response time.


Presentation Layer
User Interface Component
Profile Management API
Subscription Management API
Meal Plan Request API
Dietary Profile Component
Meal Plan Generator Component
Validation Service
Dietary Validation Component
Validated Meal Plan
Meal Plan Review & Approval Component
Subscription & Delivery Component
Database Access
Data Layer
Meal & Subscription Database